Chapter 8

Z-Transformation

Sequence

Informally, a sequence is an ordered collection of elements where each element has a position index (e.g., 1st, 2nd, 3rd, ...), and the order matters that is changing the order gives a different sequence.
Examples of some general sequences, like

$$
\begin{aligned}
y_k &= \{1\}_{k=0}^{\infty} = \{1, 1, 1, \cdots\}, \\
y_k &= \left\{a^k\right\}_{k=0}^{\infty} = \left\{1, a, a^2, \cdots\right\}, \\
y_k &= \{k\}_{k=0}^{\infty} = \{1, 2, 3, 4, \cdots\}, \\
y_k &= \left\{k^2\right\}_{k=0}^{\infty} = \{1, 4, 9, 16, \cdots\}, \\
y_k &= \left\{ka^k\right\}_{k=0}^{\infty} = \left\{a, 2a^2, 3a^3, \cdots\right\}
\end{aligned}
$$

8.1 The Z-Transform

The $z-$transform is a mathematical tool in engineering mathematics that is used to analyze and manipulate discrete-time signals and systems. It converts a discrete-time sequence into a complex-valued function in the frequency domain (z-domain), similar to how the Laplace transform handles continuous-time signals. This transformation simplifies the analysis of digital systems and is crucial in areas such as signal processing, control theory, and digital filter design.

Definition 8.1.1. *The "z-transform" of a sequence $\{y_k\}$ is a function $Y(z)$ of a complex variable defined by*

$$Y(z) = \mathcal{Z}(y_k) = \sum_{k=0}^{\infty} \frac{y_k}{z^k},$$

and we say that the z-transform "exists" provided there is a number $R > 0$ such that $\sum_{k=0}^{\infty} \frac{y_k}{z^k}$ converges for $|z| > R$. The sequence $\{y_k\}$ is said to be "exponentially bounded" if there is an

$M > 0$ and a $c > 1$ such that

$$|y_k| \leq Mc^k$$

for $k \geq 0$.

Example 8.1.1. *Find the z-transform of the sequence $\{y_k = 1\}$.*

$$\begin{aligned} Y(z) = \mathcal{Z}(1) &= \sum_{k=0}^{\infty} \frac{1}{z^k} \\ &= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \\ &= 1 + z^{-1} + (z^{-1})^2 + (z^{-1})^3 + \cdots \\ &= \frac{1}{1 - z^{-1}} \\ &= \frac{z}{z-1}, \quad |z| > 1. \end{aligned}$$

Example 8.1.2. *Find the z-transform of the sequence $\left\{x_k = a^k\right\}$.*

$$\begin{aligned} X(z) = \mathcal{Z}(a^k) &= \sum_{k=0}^{\infty} \frac{a^k}{z^k} \\ &= a^0 + \frac{a^1}{z} + \frac{a^2}{z^2} + \frac{a^3}{z^3} + \cdots \\ &= 1 + \left(\frac{a}{z}\right) + \left(\frac{a}{z}\right)^2 + \left(\frac{a}{z}\right)^3 + \cdots \\ &= \frac{1}{1 - \frac{a}{z}} \\ &= \frac{z}{z-a}, \quad |z| > |a|. \end{aligned}$$

Example 8.1.3. *Find the z-transform of* $\{y_k = k\}_{k=0}^{\infty}$.

$$\begin{aligned}
Y(z) = \mathcal{Z}(k) &= \sum_{k=0}^{\infty} \frac{k}{z^k} \\
&= 0 + \frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + \cdots \\
&= \sum_{k=0}^{\infty} \frac{k+1}{z^{k+1}} \\
&= \frac{1}{z} \sum_{k=0}^{\infty} \frac{k+1}{z^k} \\
&= \frac{1}{z} Y(z) + \frac{1}{z} \mathcal{Z}(1), \quad |z| > 1 \\
&= \frac{1}{z} Y(z) + \frac{1}{z} \cdot \frac{z}{z-1} \\
&= \frac{1}{z} Y(z) + \frac{1}{z-1} \\
\Rightarrow \quad \left(1 - \frac{1}{z}\right) Y(z) &= \frac{1}{z-1} \\
\Rightarrow \quad \frac{z-1}{z} Y(z) &= \frac{1}{z-1} \\
\Rightarrow \quad Y(z) &= \frac{z}{(z-1)^2}, \quad |z| > 1.
\end{aligned}$$

Theorem 8.1.1 (Linearity Theorem)**.** *If* a *and* b *are constants, then*

$$\mathcal{Z}(ax_k + by_k) = a\mathcal{Z}(x_k) + b\mathcal{Z}(y_k)$$

for those z *in the common domain of* $X(z)$ *and* $Y(z)$.

Example 8.1.4. *Find the z-transform of* $\{\sin ak\}_{k=0}^{\infty}$.

$$\begin{aligned}
\mathcal{Z}(\sin ak) &= Z\left(\frac{1}{2i}e^{iak} - \frac{1}{2i}e^{-iak}\right) \\
&= \frac{1}{2i}\mathcal{Z}(e^{iak}) - \frac{1}{2i}\mathcal{Z}(e^{-iak}) \\
&= \frac{1}{2i}\sum_{k=0}^{\infty}\frac{e^{iak}}{z^k} - \frac{1}{2i}\sum_{k=0}^{\infty}\frac{e^{-iak}}{z^k} \\
&= \frac{1}{2i}\left(1 + \frac{e^{ia}}{z} + \frac{e^{2ia}}{z^2} + \cdots\right) - \frac{1}{2i}\left(1 + \frac{e^{-ia}}{z} + \frac{e^{-2ia}}{z^2} + \cdots\right) \\
&= \frac{1}{2i}\left(\sum_{k=0}^{\infty}\left(\frac{e^{ia}}{z}\right)^k - \sum_{k=0}^{\infty}\left(\frac{e^{-ia}}{z}\right)^k\right) \\
&= \frac{1}{2i}\left(\frac{1}{1-\frac{e^{ia}}{z}} - \frac{1}{1-\frac{e^{-ia}}{z}}\right) \\
&= \frac{1}{2i}\left(\frac{z}{z-e^{ia}} - \frac{z}{z-e^{-ia}}\right) \\
&= \frac{z^2 - ze^{-ia} - z^2 + ze^{ia}}{2i\left[z^2 - (e^{ia}+e^{-ia})z + 1\right]} \\
&= \frac{z(e^{ia} - e^{-ia})}{2i\left[z^2 - 2z\cos a + 1\right]} \\
&= \frac{z\sin a}{z^2 - 2z\cos a + 1}
\end{aligned}$$

Similarly, one can show that

$$\begin{aligned}
\mathcal{Z}(\cos ak) &= Z\left(\frac{1}{2}e^{iak} + \frac{1}{2}e^{-iak}\right) \\
&= \frac{1}{2}\left(\frac{z}{z-e^{ia}} + \frac{z}{z-e^{-ia}}\right) \\
&= \frac{z(z-e^{-ia}) + z(z-e^{ia})}{2\left[z^2 - 2z\cos a + 1\right]} \\
&= \frac{z^2 - z\cos a}{z^2 - 2z\cos a + 1}
\end{aligned}$$

Theorem 8.1.2. *If* $Y(z) = \mathcal{Z}(y_k)$ *for* $|z| > r$, *then*

$$Z\left((k+n-1)^n y_k\right) = (-1)^n z^n \frac{d^n Y}{dz^n}(z)$$

for $|z| > r$.

Example 8.1.5. *Find the z-transform of the sequence* $\left\{ka^k\right\}_{k=0}^{\infty}$.

$$\begin{aligned}
\mathcal{Z}(ka^k) &= \sum_{k=0}^{\infty} \frac{ka^k}{z^k} \\
&= \sum_{k=0}^{\infty} k\left(\frac{a}{z}\right)^k \\
&= -z\frac{d}{dz}\left(\sum_{k=0}^{\infty}\left(\frac{a}{z}\right)^k\right) \\
&= -z\frac{d}{dz}\left(\frac{1}{1-\frac{a}{z}}\right), \quad |z| > |a| \\
&= -z\cdot\frac{d}{dz}\left(\frac{z}{z-a}\right) \\
&= -z\cdot\left(\frac{(z-a)\cdot 1 - z\cdot 1}{(z-a)^2}\right) \\
&= -z\cdot\left(\frac{z-a-z}{(z-a)^2}\right) \\
&= -z\cdot\left(\frac{-a}{(z-a)^2}\right) \\
&= \frac{az}{(z-a)^2}
\end{aligned}$$

Example 8.1.6. *Find the z-transform of the sequence* $\{k^2\}_{k=0}^{\infty}$.

$$\begin{aligned}
\mathcal{Z}(k^2) &= \sum_{k=0}^{\infty} \frac{k^2}{z^k} \\
&= \mathcal{Z}(k \cdot k) \\
&= -z\frac{d}{dz}\mathcal{Z}(k) \\
&= -z\frac{d}{dz}\left(\frac{z}{(z-1)^2}\right) \\
&= -z \cdot \left(\frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1)}{(z-1)^4}\right) \\
&= -z \cdot \left(\frac{(z-1)^2 - 2\mathcal{Z}(z-1)}{(z-1)^4}\right) \\
&= -z \cdot \left(\frac{z^2 - 2z + 1 - 2z^2 + 2z}{(z-1)^4}\right) \\
&= -z \cdot \left(\frac{-z^2+1}{(z-1)^4}\right) \\
&= \frac{\mathcal{Z}(z^2-1)}{(z-1)^4} \\
&= \frac{\mathcal{Z}(z+1)(z-1)}{(z-1)^4} \\
&= \frac{\mathcal{Z}(z+1)}{(z-1)^3}
\end{aligned}$$

Definition 8.1.2. *Define the unit step sequence* $u(n)$ *by*

$$u_k(n) = \begin{cases} 0, & 0 \le k \le n-1 \\ 1, & n \le k. \end{cases}$$

Note that the unit step sequence has a single "step" of unit height located at $k = n$. The following result is known as a "shifting theorem."

Example 8.1.7. *Find the z-transform of the shifted unit step sequence* $\{u_k(n)\}_{k=0}^{\infty}$, *where*

$$u_k(n) = \begin{cases} 1, & k \ge n \\ 0, & k < n \end{cases}$$

$$
\begin{aligned}
\mathcal{Z}(u_k(n)) &= \sum_{k=0}^{\infty} \frac{u_k(n)}{z^k} \\
&= \frac{x_0(n)}{z^0} + \frac{u_1(n)}{z^1} + \frac{u_2(n)}{z^2} + \cdots + \frac{u_n(n)}{z^n} + \frac{u_{n+1}(n)}{z^{n+1}} + \cdots \\
&= \underbrace{\frac{\cancelto{0}{x_0(n)}}{z^0} + \frac{\cancelto{0}{u_1(n)}}{z^1} + \frac{\cancelto{0}{u_2(n)}}{z^2} + \cdots + \frac{\cancelto{0}{u_{n-1}(n)}}{z^{n-1}}}_{=0, as\ k<n} + \frac{u_n(n)}{z^n} + \frac{u_{n+1}(n)}{z^{n+1}} + \cdots \\
&= \sum_{k=n}^{\infty} \frac{1}{z^k} \\
&= \frac{1}{z^n} + \frac{1}{z^{n+1}} + \frac{1}{z^{n+2}} + \cdots \\
&= \frac{1}{z^n}\left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right) \\
&= \frac{1}{z^n} \cdot \frac{1}{1 - \frac{1}{z}}, \quad |z| > 1 \\
&= \frac{1}{z^n} \cdot \frac{z}{z-1} \\
&= \frac{z^{1-n}}{z-1}
\end{aligned}
$$

Definition 8.1.3. *We now define the unit impulse sequence $\delta(n)$, $n \geq 1$, by*

$$\delta_k(n) = \begin{cases} 1, & k = n \\ 0, & k \neq n. \end{cases}$$

Example 8.1.8. *Find the z-transform of the shifted delta sequence $\{\delta_k(n)\}_{k=0}^{\infty}$, where*

$$\delta_k(n) = \begin{cases} 1, & k = n \\ 0, & k \neq n \end{cases}$$

$$
\begin{aligned}
\mathcal{Z}(\delta_k(n)) &= \sum_{k=0}^{\infty} \frac{\delta_k(n)}{z^k} \\
&= \frac{\delta_0(n)}{z^0} + \frac{\delta_1(n)}{z^1} + \frac{\delta_2(n)}{z^2} + \cdots + \frac{\delta_n(n)}{z^n} + \frac{\delta_{n+1}(n)}{z^{n+1}} + \cdots \\
&= \underbrace{\frac{\cancelto{0}{\delta_0(n)}}{z^0} + \frac{\cancelto{0}{\delta_1(n)}}{z^1} + \cdots + \frac{\cancelto{0}{\delta_{n-1}(n)}}{z^{n-1}}}_{=0, since\ k<n} + \frac{\delta_n(n)}{z^n} + \underbrace{\frac{\cancelto{0}{\delta_{n+1}(n)}}{z^{n+1}} + \cdots}_{=0, since\ k>n} \\
&= \frac{1}{z^n}
\end{aligned}
$$

Formulas Related to $Z-$transform:

Sequence	Z-transform	ROC
1	$\frac{z}{z-1}$	$\|z\|>1$
a^k	$\frac{z}{z-a}$	$\|z\|>a$
k	$\frac{z}{(z-1)^2}$	$\|z\|>1$
k^2	$\frac{\mathcal{Z}(z+1)}{(z-1)^3}$	$\|z\|>1$
ky_k	$-zY'(z)$	$---$
ka^k	$\frac{az}{(z-a)^2}$	$\|z\|>a$
$\sin(ak)$	$\frac{z\sin a}{z^2-2z\cos a+1}$	$---$
$\cos(ak)$	$\frac{z^2-z\cos a}{z^2-2z\cos a+1}$	$---$
$\sin\left(\frac{\pi}{2}k\right)$	$\frac{z}{z^2+1}$	$---$
$\cos\left(\frac{\pi}{2}k\right)$	$\frac{z^2}{z^2+1}$	$---$
$\sinh(ak)$	$\frac{z\sinh a}{z^2-2z\cosh a+1}$	$---$
$\cosh(ak)$	$\frac{z^2-z\cosh a}{z^2-2z\cosh a+1}$	$---$
$\delta_k(n)$	$\frac{1}{z^n}$	$---$
$u_k(n)$	$\frac{z^{1-n}}{z-1}$	$---$
$a^k y_k$	$Y\left(\frac{z}{a}\right)$	$---$
y_{k+1}	$zY(z)-zy_0$	$---$
y_{k+2}	$z^2Y(z)-z^2y_0-zy_1$	$---$

Example 8.1.9. $Z\left\{\left(\frac{3}{2}\right)^k+5u_k(4)+3^{k+1}+\delta_k(-3)\right\}$

$$= \frac{z}{z-\frac{3}{2}}+\frac{5z^{1-4}}{z-1}+3\frac{z}{z-3}+\frac{1}{z^{-3}}$$

$$= \frac{2z}{2z-3}+\frac{5z^{-3}}{z-1}+3\frac{z}{z-3}+z^3$$

Example 8.1.10. $Z\{6k + 5\cos 3k + 2k^2 + 3u_k(-2)\}$

$$= \frac{6z}{(z-1)^2} + 5\frac{z^2 - z\cos 3}{z^2 - 2z\cos 3 + 1} + \frac{2z(z+1)}{(z-1)^3} + \frac{3z^3}{z-1}$$

Example 8.1.11. *Find* $\mathcal{Z}(y_k)$ *if* $y_k = 2, 0 \le k \le 99, y_k = 5, 100 \le k$.

$$\begin{aligned}\mathcal{Z}(y_k) &= 2\mathcal{Z}(1) + 3\mathcal{Z}(u_k(100)) \\ &= \frac{2z}{z-1} + \frac{3z^{-99}}{z-1} \\ &= \frac{2z^{100} + 3}{z^{99}(z-1)}.\end{aligned}$$

Theorem 8.1.3. *If* $\mathcal{Z}(y_k) = Y(z)$ *for* $|z| > r$, *then for constants* $a \neq 0$,

$$Z\left(a^k y_k\right) = Y\left(\frac{z}{a}\right)$$

for $|z| > r|a|$.

Example 8.1.12. *Find* $\mathcal{Z}(3^k \sin 4k)$.

$$\mathcal{Z}(3^k \sin 4k) = \mathcal{Z}(\sin 4k)\Big|_{\frac{z}{3}}$$

$$\begin{aligned}&= \frac{\frac{z}{3}\sin 4}{\frac{z^2}{9} - 2(\cos 4)\frac{z}{3} + 1} \\ &= \frac{3z\sin 4}{z^2 - 6z\cos 4 + 9}.\end{aligned}$$

Example 8.1.13. *Find* $\mathcal{Z}(k\cos\frac{k\pi}{2})$.

$$\begin{aligned}&= -z\frac{d}{dz}\frac{z^2 - z\cos\frac{\pi}{2}}{z^2 - 2z\cos\frac{\pi}{2} + 1} \\ &= -z\frac{d}{dz}\frac{z^2}{z^2+1} \\ &= -z\frac{(z^2+1)2z - z^2\cdot 2z}{(z^2+1)^2} \\ &= -z\frac{2z}{(z^2+1)^2} \\ &= -\frac{2z^2}{(z^2+1)^2}\end{aligned}$$

8.2 Inverse Z-Transform

The inverse transformation, from complex frequency domain back to discrete time domain is called the inverse $z-$transform. It helps solve difference equations, which are a key component of numerous scientific and engineering problems, including linear time-invariant systems.

The inverse Z-Transform of $Y(z)$ is given by:

$$Z^{-1}\{Y(z)\} = y_k$$

Common Formulas Related to Inverse $z-$Transform:

$$Z^{-1}\left\{\frac{z}{z-1}\right\} = 1$$
$$Z^{-1}\left\{\frac{z}{z-a}\right\} = a^k$$
$$Z^{-1}\left\{\frac{z}{(z-1)^2}\right\} = k$$
$$Z^{-1}\left\{\frac{az}{(z-a)^2}\right\} = ka^k$$
$$Z^{-1}\left\{\frac{z^{1-m}}{z-1}\right\} = u_k(m)$$
$$Z^{-1}\left\{\frac{z\sin a}{z^2-2z\cos a+1}\right\} = \sin(ak)$$
$$Z^{-1}\left\{\frac{z^2-z\cos a}{z^2-2z\cos a+1}\right\} = \cos(ak)$$
$$Z^{-1}\left\{\frac{z}{z^2+1}\right\} = \sin\left(\frac{\pi}{2}k\right)$$
$$Z^{-1}\left\{\frac{z^2}{z^2+1}\right\} = \cos\left(\frac{\pi}{2}k\right)$$

Example 8.2.1. *Evaluate* $Z^{-1}\left\{\frac{z}{z-2}+\frac{z}{(z-3)^2}+z^{-4}\right\}$

$$Z^{-1}\left\{\frac{z}{z-2}+\frac{z}{(z-3)^2}+z^{-4}\right\}$$
$$= 2^k + \frac{1}{3}k3^k + \delta_k(4)$$

Example 8.2.2. *Find the inverse* $z-$ *transform of*

$$Y(z) = \frac{2z^2-3z}{z^2-3z-4}$$

$$\Rightarrow Y(z) = \frac{\mathcal{Z}(2z-3)}{z^2-4z+z-4}$$
$$\Rightarrow \frac{Y(z)}{z} = \frac{2z-3}{(z-4)(z+1)}$$
$$\Rightarrow \frac{Y(z)}{z} = \frac{1}{z-4} + \frac{1}{z+1} \textit{(by using cover up rule)}$$
$$\Rightarrow Y(z) = \frac{z}{z-4} + \frac{z}{z+1}$$

Applying inverse $z-$ transform, we get

$$y_k = 4^k + (-1)^k$$

Example 8.2.3. *Find the sequence whose $z-$transform is $Y(z) = \frac{z^2+3z}{(z-3)^2}$*

$$Y(z) = \frac{z^2+3z}{(z-3)^2}$$
$$\Rightarrow Y(z) = \frac{\mathcal{Z}(z+3)}{(z-3)^2}$$
$$\Rightarrow \frac{Y(z)}{z} = \frac{z+3}{(z-3)^2}$$
$$\Rightarrow \frac{Y(z)}{z} = \frac{1}{z-3} + \frac{6}{(z-3)^2} \textit{(by using partial fraction)}$$
$$\Rightarrow Y(z) = \frac{z}{z-3} + \frac{6z}{(z-3)^2}$$

Applying inverse $z-$ transform, we get

$$y_k = 3^k + 2k3^k$$

8.3 Solution of Difference Equations using $\mathcal{Z}$-Transform

Definition 8.3.1. *(Difference Equation) A difference equation is a mathematical equation that relates the values of a sequence at different points. It's like a discrete version of a differential equation and is used when the independent variable (like time) can only take on discrete values.*

A first order difference equation is a recursively defined sequence in the form

$$y_{k+1} = f(k, y_k), k = 0, 1, 2, \ldots$$

Theorem 8.3.1. *For n a positive integer*

$$\mathcal{Z}(y_{k+n}) = z^n \mathcal{Z}(y_k) - \sum_{m=0}^{n-1} y_m z^{n-m},$$
$$\mathcal{Z}(y_{k-n}u_k(n)) = z^{-n}\mathcal{Z}(y_k).$$

Example 8.3.1. *Solve the following initial value problem using z-transforms:*

$$y_{k+1} - 3y_k = 4,$$

$$y_0 = 1.$$

Taking the z-transform of both sides of the difference equation, we have

$$zY(z) - zy_0 - 3Y(z) = 4\frac{z}{z-1}$$

$$(z-3)Y(z) = z + \frac{4z}{z-1} = \frac{z^2+3z}{z-1}$$

$$Y(z) = z \cdot \frac{z+3}{(z-1)(z-3)}$$

$$= z\left[\frac{-2}{z-1} + \frac{3}{z-3}\right]$$

$$= -2\frac{z}{z-1} + 3\frac{z}{z-3}.$$

By applying inverse z-transform, we find the solution

$$y_k = -2 + 3^{k+1}.$$

Example 8.3.2. *Solve the initial value problem*

$$y_{k+1} - 3y_k = 3^k,$$

$$y_0 = 2.$$

Taking the z-transform of both sides of the difference equation, we have

$$zY(z) - 2z - 3Y(z) = \frac{z}{z-3}$$

$$(z-3)Y(z) = 2z + \frac{z}{z-3} = \frac{2z^2-5z}{z-3}$$

$$Y(z) = z\frac{2z-5}{(z-3)^2}$$

$$= z\left[\frac{2}{z-3} + \frac{1}{(z-3)^2}\right]$$

$$= 2\frac{z}{z-3} + \frac{1}{3}\frac{3z}{(z-3)^2}.$$

Then

$$y_k = 2 \cdot 3^k + \frac{1}{3}k3^k.$$

Example 8.3.3. *Solve the second order initial value problem*

$$y_{k+2} + y_k = 10 \cdot 3^k$$
$$y_0 = 0, \quad y_1 = 0.$$

By the shifting theorem,

$$z^2\mathcal{Z}(y_k) - y_0 z^2 - y_1 z + \mathcal{Z}(y_k) = \frac{10z}{z-3}$$
$$(z^2+1)\mathcal{Z}(y_k) = \frac{10z}{z-3},$$

so we have

$$\begin{aligned}\mathcal{Z}(y_k) &= \frac{10z}{(z-3)(z^2+1)} \\ &= z\left[\frac{A}{z-3} + \frac{Bz+C}{z^2+1}\right] \\ &= z\left[\frac{1}{z-3} - \frac{z+3}{z^2+1}\right] \\ &= \frac{z}{z-3} - \frac{z^2}{z^2+1} - \frac{3z}{z^2+1} \\ &= \frac{z}{z-3} - \frac{z^2 - z\cos\frac{\pi}{2}}{z^2 - 2z\cos\frac{\pi}{2} + 1} - \frac{3z\sin\frac{\pi}{2}}{z^2 - 2z\cos\frac{\pi}{2} + 1}.\end{aligned}$$

Hence

$$y_k = 3^k - \cos\left(\frac{\pi}{2}k\right) - 3\sin\left(\frac{\pi}{2}k\right).$$

Example 8.3.4. *Solve the initial value problem*

$$y_{k+1} - 2y_k = 3\delta_k(4), \quad y_0 = 1.$$

Taking the z-transform of both sides, we have

$$zY(z) - z - 2Y(z) = \frac{3}{z^4}$$
$$(z-2)Y(z) = z + \frac{3}{z^4}$$
$$\begin{aligned}Y(z) &= \frac{z}{z-2} + \frac{3}{z^4(z-2)} \\ &= \frac{z}{z-2} + 3z^{-5}\frac{z}{z-2}.\end{aligned}$$

By taking the inverse z-transform, we get

$$y_k = 2^k + 3 \cdot 2^{k-5}u_k(5).$$

We can also write this in the form

$$y_k = \begin{cases} 2^k, & 0 \le k \le 4 \\ 2^k + 3 \cdot 2^{k-5}, & k \ge 5. \end{cases}$$

Example 8.3.5. *Solve the following system of difference equations*

$$\begin{cases} x_{k+1} - 3y_k = 0 \\ 4x_k - 3y_{k+1} = 0 \end{cases} \qquad \textit{subject to} \quad x_0 = 3, \quad y_0 = -2$$

Solution: Applying the z-transform on both of the difference equations, we have

$$\begin{cases} zX(z) - zx_0 - 3Y(z) = 0 \\ 4X(z) - 3zY(z) + 3zy_0 = 0 \end{cases}$$

Using initial conditions, we obtain

$$\implies \begin{cases} zX(z) - 3z - 3Y(z) = 0 \\ 4X(z) - 3zY(z) - 6z = 0 \end{cases}$$

$$\implies \begin{cases} zX(z) - 3Y(z) = 3z \\ 4X(z) - 3zY(z) = 6z \end{cases}$$

Now, let's define

$$D = \begin{vmatrix} z & -3 \\ 4 & -3z \end{vmatrix} = -3z^2 + 12 = -3(z^2 - 4)$$

$$D_X = \begin{vmatrix} 3z & -3 \\ 6z & -3z \end{vmatrix} = -9z^2 + 18z = -9z(z - 2)$$

$$D_Y = \begin{vmatrix} z & 3z \\ 4 & 6z \end{vmatrix} = 6z^2 - 12z = 6z(z - 2)$$

According to Cramer's rule, we can write

$$X(z) = \frac{D_X}{D} = \frac{3z}{z + 2}$$

$$Y(z) = \frac{D_Y}{D} = \frac{-2z}{z + 2}$$

Therefore, by inverse $\mathcal{Z}-$ transform, the solution of the system will be

$$x_k = 3(-2)^k$$

and

$$y_k = -2(-2)^k = (-2)^{k+1}$$

Example 8.3.6. *Solve the system*

$$x_{k+1} - y_k = 3k \cdot 3^k,$$

$$x_k + y_{k+1} - 3y_k = k \cdot 3^k,$$
$$x_0 = 0, \quad y_0 = 3.$$

Applying the z-transform on both of the difference equations, we have

$$zX(z) - zx_0 - Y(z) = \frac{9z}{(z-3)^2},$$

$$X(z) + zY(z) - zy_0 - 3Y(z) = \frac{3z}{(z-3)^2},$$

or

$$zX(z) - Y(z) = \frac{9z}{(z-3)^2},$$
$$X(z) + (z-3)Y(z) = 3z + \frac{3z}{(z-3)^2}.$$

Multiplying both sides of the above equation by $(z-3)^2$ *and adding, we get*

$$z(z-3)^2X(z) - (z-3)^2Y(z) = 9z,$$
$$(z-3)^2X(z) + (z-3)^3Y(z) = 3z(z-3)^2 + 3z.$$

or

$$z(z-3)^2X(z) - (z-3)^2Y(z) = 9z,$$
$$(z-3)^2X(z) + (z-3)^3Y(z) = 3z\left\{(z-3)^2 + 1\right\}.$$

Now, let's define

$$D = \begin{vmatrix} z(z-3)^2 & -(z-3)^2 \\ (z-3)^2 & (z-3)^3 \end{vmatrix} = z(z-3)^5 + (z-3)^4 = (z-3)^4(z(z-3)+1) = (z-3)^4(z^2-3z+1)$$

$$D_X = \begin{vmatrix} 9z & -(z-3)^2 \\ 3z((z-3)^2+1) & (z-3)^3 \end{vmatrix} = 9z(z-3)^3 + 3z(z-3)^2((z-3)^2+1) = 3z(z-3)^2(3(z-3)+(z-3)^2+1)$$

$$D_X = 3z(z-3)^2(3z - 9 + z^2 - 6z + 9 + 1) = 3z(z-3)^2(z^2 - 3z + 1)$$

$$X = \frac{D_X}{D} = \frac{3z(z-3)^2(z^2-3z+1)}{(z-3)^4(z^2-3z+1)} = \frac{3z}{(z-3)^2}$$

and by applying inverse $z-$*transform, we obtain*

$$x_k = k3^k.$$

Furthermore,

$$y_k = x_{k+1} - 3k3^k = (k+1)3^{k+1} - 3k3^k = k3 \cdot 3^k + 3 \cdot 3^k - 3k \cdot 3^k = 3^{k+1}.$$

8.4 Exercises

1. Find the $\mathcal{Z}$-transform of each of the following:

   (a) $y_k = 2 + 3k$
   (b) $y_k = (4 + k)^2$
   (c) $y_k = 2^{k-2}$
   (d) $y_k = \left(-\frac{1}{2}\right)^k$
   (e) $y_k = \delta_k(2)$
   (f) $y_k = \delta_k(-3)$
   (g) $y_k = u_k(3)$
   (h) $y_k = u_k(-5)$
   (i) $y_k = 3^k \cos 2k$
   (j) $y_k = 2^k u_k(n)$
   (k) $y_k = k3^{-k}$
   (l) $y_k = 5^k \cos\left(\frac{\pi}{2}k\right)$
   (m) $y_k = 5^{-k} \sin\left(\frac{\pi}{2}k\right)$
   (n) $y_k = 3^k \cos\left(\frac{\pi}{2}\right)$
   (o) $y_k = k \cos\left(\frac{\pi}{2}k\right)$
   (p) $y_k = \sin(2k - 3)$
   (q) $y_k = ku_k(7)$

2. Evaluate the following $\mathcal{Z}-$ transformations:

   (a) $\mathcal{Z}\left\{(-3)^k + \cos 5k + 4k^2 + 7u_k(5) + \delta_k(2)\right\}$.
   (b) $\mathcal{Z}\left\{7 - (\frac{-5}{2})^k + 3\cos \pi k + 7k^2 + \frac{3}{2}u_k(5)\right\}$.
   (c) $\mathcal{Z}\left\{k5^k + 2^{k+4} + 3k \sin\left(\frac{\pi}{2}k\right) + 3^k\delta_k(-6)\right\}$.
   (d) $\mathcal{Z}\left\{7k + 2^k \cos\left(\frac{\pi}{2}k\right) + ku_k(3)\right\}$.
   (e) $\mathcal{Z}\left\{(3)^{2-k} + k\delta_k(4) + 5^k \sin\left(\frac{\pi}{2}k\right)\right\}$.

3. Derive the formulae for $\mathcal{Z}(k^2)$, $\mathcal{Z}(k^3)$, $\mathcal{Z}(\cosh ak)$ and $\mathcal{Z}(\cos ak)$.

4. Evaluate the following inverse $\mathcal{Z}-$ transformations:

   (a) $\mathcal{Z}^{-1}\left\{\frac{2z}{z-3} + \frac{3z}{(z-4)^2} + 4z^{-3}\right\}$.
   (b) $\mathcal{Z}^{-1}\left\{\frac{2z}{(z-1)^2} + \frac{1}{z^4(z-1)} + \frac{2z^2}{z^2+1}\right\}$.
   (c) $\mathcal{Z}^{-1}\left\{\frac{6z}{2z+4} + \frac{6z}{z^2-6z+9} + \frac{3z}{z^2+1} + \frac{z^{-3}}{z-1}\right\}$.

5. Find the sequences whose $\mathcal{Z}$-transforms are

   (a) $Y(z) = \frac{2z^2-3z}{z^2-3z-4}$.
   (b) $X(z) = \frac{3z^2-4z}{z^2-3z+2}$.
   (c) $Y(z) = \frac{2z^2+z}{(z-1)^2}$.
   (d) $Y(z) = \frac{z}{2z^2-2\sqrt{2}z+2}$.
   (e) $X(z) = \frac{2z^2-z}{2z^2-2z+2}$.
   (f) $Y(z) = \frac{z^2+3z}{(z-3)^2}$.

(g) $W(z) = \frac{3z^2+5}{z^4}$.

6. Derive the formula for $\mathcal{Z}(\delta_k(n))$ by expressing $\delta_k(n)$ in terms of step functions.

7. Solve the following first order initial value problems using z-transforms.

   (a) $y_{k+1} - 3y_k = 4^k, \quad y_0 = 0.$

   (b) $y_{k+1} + 4y_k = 10, \quad y_0 = 3.$

   (c) $y_{k+1} - 5y_k = 5^{k+1}, \quad y_0 = 0.$

   (d) $y_{k+1} - 2y_k = 3 \cdot 2^k, \quad y_0 = 3.$

   (e) $y_{k+1} - y_k = 2 \sin \frac{\pi}{2} k, \quad y_0 = 1.$

   (f) $y_{k+1} + 3y_k = 4\delta_k(2), \quad y_0 = 2.$

8. Solve the following second order initial value problems using z-transforms:

   (a) $y_{k+2} - 5y_{k+1} + 6y_k = 0, \quad y_0 = 1, \quad y_1 = 0.$

   (b) $y_{k+2} - y_{k+1} - 6y_k = 0, \quad y_0 = 5, \quad y_1 = -5.$

   (c) $y_{k+2} - 8y_{k+1} + 16y_k = 0, \quad y_0 = 0, \quad y_1 = 4.$

   (d) $y_{k+2} - y_k = 16 \cdot 3^k, \quad y_0 = 2, \quad y_1 = 6.$

   (e) $y_{k+2} - 3y_{k+1} + 2y_k = u_k(4), \quad y_0 = 0, \quad y_1 = 0.$

9. Solve the following systems using z-transforms:

   (a) $x_{k+1} + y_k = x_k, \quad x_k + y_{k+1} = y_k, x_0 = 1, \quad y_0 = -1.$

   (b) $x_{k+1} - 2y_k = 2 \cdot 4^k, -4x_k + y_{k+1} = 4^{k+1}, x_0 = 2, \quad y_0 = 3.$

   (c) $x_{k+1} - y_k = 0, x_k + y_{k+1} = 0, x_0 = 0, \quad y_0 = 1.$

   (d) $x_{k+1} - y_k = 2k, -x_k + y_{k+1} = 2k + 2, x_0 = 0, \quad y_0 = 1.$

   (e) $x_{k+1} - y_k = -1, -x_k + y_{k+1} = 3, x_0 = 0, \quad y_0 = 2.$